ScreenSense: Kids' Screentime Visualization

Milestone 4: Report and Presentation

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Analysis for Kids

1. Overall Screen Time Levels

Urban and rural kids have **very similar average screen time**.

- Urban: around **4.4 hours/day**
- Rural: around **4.3 hours/day**

What this really means:
Access is different, but behaviour is almost the same. Rural kids are not "protected" from high screen time — they are catching up quickly.

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2. Health Impact Patterns

Urban kids show **much higher counts** across almost all health issues:

- Eye strain
- Anxiety
- Poor sleep
- Obesity risk
- Potential long-term risk

Interpretation:
Urban kids have higher exposure to devices, so health issues appear more frequently. Rural kids report fewer health problems, likely due to slightly lower device intensity and lower device diversity.

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3. Device Usage Differences

The data shows bigger device variety in urban areas:

- Urban kids use **smartphones and TVs heavily**, with strong counts.

- Rural kids show **lower usage overall**, especially for laptops and tablets.

Meaning:
Urban kids have wider access to personal devices.
Rural kids rely on a narrower set of devices (mostly shared ones like TVs).

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4. Screen Time Category Patterns

Age X Location Cohort show similar proportions of:

- Moderate usage
- High usage
- Very high usage

across both groups, with no major rural–urban gap in.

What this tells us:
Even though rural kids use fewer devices, their **screen time hours** end up close to urban kids.

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5. Age-Based Differences: Urban vs Rural

From the Age × Location Cohort table:

- Kids (8–11) show **lower averages**, but even here urban kids are a little higher than rural kids.

Meaning:
Digital behaviour increases with age regardless of location.
Urban kids show a small but consistent upward shift compared to rural kids.

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6. Day-Type Differences (Weekend vs Weekday)

Weekend usage spikes strongly for both groups.

- Weekend screen time ≈ **6 hours**
- Weekday screen time ≈ **3.5 hours**

Interpretation:
Both urban and rural kids **double their usage on weekends**, showing the same behavioural rhythm.

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7. Educational vs Recreational Ratio

Urban kids show:

- More **balanced** or **recreational** usage
- Less "mostly educational" usage

Rural kids show:

- Similar pattern, but **overall lower counts** in every category.

Insight:
Urban kids use screens for a broader mix of entertainment and learning.
Rural kids skew slightly more to recreational use because educational digital resources are less accessible.

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8. Exceeded Recommended Limit

Urban and rural kids both exceed screen-time limits at very high levels.

Visuals show that most children exceed recommended limits regardless of their location group.

Meaning:
Location is not a protective factor.

Analysis for Pre-Teens

1. Overall Screen Time

From the Age × Location Cohort table:

- **Pre-Teens – Rural: 4.54 hrs/day**
- **Pre-Teens – Urban: 4.52 hrs/day**

Meaning:
Urban and rural pre-teens behave almost identically in terms of total hours. The location doesn't influence how long they stay on screens.

This is one of the strongest findings: **location does not change pre-teen behaviour.**

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2. Device Usage Patterns

Device × Location visuals show clear trends:

- Urban children show **higher total device usage**, especially smartphones, TVs, and laptops.
- Rural children use **fewer device types**, mainly smartphones and TVs.

Interpretation for pre-teens:
Pre-teens in urban areas have wider device access → more variety in screen habits.
Pre-teens in rural areas rely on fewer devices → mostly smartphones and TVs.

But despite the difference in access, **their total screen hours remain the same** (around 4.5 hours).

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3. Screen Time Category Behaviour (Low, Moderate, High, Very High)

Pre-teens overall show:

- Very **high representation in Moderate + High usage groups**
- Very **small counts in Very High usage**
- Almost no Low usage among urban or rural pre-teens

This comes from the Screen Time Category distribution table and visuals.

Because the rural/urban pre-teen averages are nearly identical, both groups fall into the same usage brackets.

Meaning:
Regardless of location, pre-teens tend to be in the "middle to high" usage zone.
Location does not shift their category significantly.

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4. Weekend vs Weekday Patterns

Your weekend vs weekday chart shows:

- Weekend screen time ≈ **6.1 hours**
- Weekday screen time ≈ **3.5 hours**

Since urban and rural pre-teens have nearly identical averages (4.54 vs 4.52), this weekend spike pattern applies equally to both.

Meaning:
Both urban and rural pre-teens double their usage on weekends.
This is age-driven, not location-driven.

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5. Educational vs Recreational Usage (EduRec Category)

Across locations:

- **Balanced usage** is the most common pattern.
- **Mostly recreational** is the next most common.
- **Mostly educational** is very rare for both urban and rural.

Interpretation for pre-teens:
Urban and rural pre-teens both:

- use screens mainly for entertainment or balanced purposes
- rarely use devices for learning

This pattern is consistent across the entire dataset.

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6. Health Impact Differences

This is where urban vs rural pre-teens show the clearest distinction.

From Health Impacts by Location:

Urban children report:

- far more **eye strain**

- far more **poor sleep**
- far more **anxiety**
- far more **potential risk**
- far more **obesity risk**

Rural children show **much lower counts across all categories**.

For pre-teens specifically:
Since pre-teens in general report high health-impact counts, urban pre-teens will appear heavily in these numbers.

Meaning:
Urban pre-teens suffer more health effects → likely due to greater device variety and more late-night usage.
Rural pre-teens report fewer issues → but not because they use screens less; they use fewer **types** of devices and may have fewer long-duration device sessions.

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7. Cohort Risk Profile (Pre-Teens in different groups)

From Age × ScreenTime Cohort averages:

- Pre-Teens – Very High Usage = **8.41 hrs/day**
- Pre-Teens – High Usage = **5.78 hrs/day**
- Pre-Teens – Moderate Usage = **3.94 hrs/day**
- Pre-Teens – Low Usage = **1.55 hrs/day**

Since urban and rural averages are almost the same, both groups of pre-teens fall into the **same risk brackets**.

Urban pre-teens tend to enter high-risk categories more often because:

- They have more device options
- They show stronger late-night use (supported by higher health impacts)

Analysis for Teens

1. Overall Screen Time (Urban Teens vs Rural Teens)

From the Age × Location Cohort table:

- **Teens – Rural: 4.52 hrs/day**
- **Teens – Urban: 4.51 hrs/day**

Meaning:
Just like pre-teens, rural and urban teens show **almost identical daily screen time**.
Location does not change screen-use intensity for teenagers.

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2. Device Usage Patterns

From the device × location visuals:

- Urban teens use **more smartphones, TVs, laptops, and tablets** overall.
- Rural teens also use smartphones and TVs heavily, but **laptop and tablet usage drops sharply**.

Interpretation:
Urban teens → wider device access → more diverse digital habits
Rural teens → rely mainly on smartphones and TVs → fewer device categories available

But **they still reach the same screen time hours**, which suggests:

- longer continuous sessions
- fewer but more intensive activities
- shared-device culture in rural settings

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3. Screen Time Category Behaviour

Using the ScreenTime Category distribution and counts:

Teens overall have:

- very high Moderate usage
- strong High usage
- very few Low usage
- some Very High usage

Since rural and urban teens have identical average screen time (4.51 vs 4.52), they fall into **the same category patterns**.

Meaning:
Whether urban or rural, teens cluster in the **3–6 hour range**, with many crossing into high-usage territory.

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4. Weekend vs Weekday Behaviour

The weekend/weekday chart applies equally because both groups share the same daily average:

- Weekend ≈ **6.1 hrs**
- Weekday ≈ **3.5 hrs**

Interpretation:
Urban and rural teens show the same rhythm:

- controlled usage during school days
- heavy spike on weekends

This is an **age-driven pattern**, not a location-driven one.

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5. Educational vs Recreational Usage

From EduRec vs Location:

Both rural and urban teens show:

- **Balanced** category as the most frequent
- **Mostly recreational** usage as the next biggest
- Very little **mostly educational** usage overall

Meaning:
Urban teens → more diverse apps, more entertainment variety
Rural teens → fewer devices but still mostly entertainment-oriented

But the ratio patterns remain almost the same.

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6. Health Impact Differences

This is where urban and rural teens truly separate.

From Health Impacts by Location:

Urban teens show much higher counts in:

- Eye Strain
- Anxiety
- Poor Sleep
- Obesity Risk
- Potential Risk

Rural teens have **much lower counts in every category**.

Interpretation:
Urban teens experience:

- longer exposure to screens
- more late-night device use
- heavier multitasking across multiple devices

Rural teens:

- have fewer devices
- report fewer health symptoms
- may have more parental or environmental restrictions

Even though screen time hours are the same, **the health stress is much higher for urban teens**.

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7. Cohort Risk Profiles

From the Age × ScreenTime Cohort table:

- Teens – Very High Usage = **8.19 hrs/day**
- Teens – High Usage = **5.79 hrs/day**
- Teens – Moderate Usage = **3.91 hrs/day**
- Teens – Low Usage = **1.50 hrs/day**

Because urban and rural teens share the same average hours, they fall into the same risk groups. However:

- Urban teens are more represented in Very High Usage (due to device availability).

- Rural teens show fewer extreme outliers.

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8. Age × Health Cohort Patterns

From the Age × Health Impact cohort table:

Teens show:

- High screen time for poor sleep
- High screen time for eye strain
- High screen time for anxiety
- All “healthy” teen groups show the **lowest averages**

Since urban teens dominate health-impact numbers, they are more present in high-risk cohorts, while rural teens appear more in the lower-risk categories.

Analysis Dimension	Urban Kids	Rural Kids	Urban Pre-Teens	Rural Pre-Teens	Urban Teens	Rural Teens
Avg Screen Time	~4.12 hrs/day	~4.04 hrs/day	~4.52 hrs/day	~4.54 hrs/day	~4.51 hrs/day	~4.52 hrs/day
Device Usage	Wide access: smartphones, TV, some laptops/tablets	Mostly smartphones and TV	High access across phones, TV, laptops	Limited device mix, mainly phones & TV	Broad access: smartphones, TV, laptops, tablets	Fewer device types overall
Screen Time Categories	Mostly Moderate, some High	Mostly Moderate, fewer High	Moderate + High dominate	Moderate + High dominate	Moderate + High dominate	Moderate + High dominate
Weekend vs Weekday	Big weekend jump	Same jump	Strong weekend spike	Same spike	Big weekend spike	Same spike
Educational vs Recreational Pattern	Mostly Balanced, some Recreational	Mostly Recreational	Balanced → Recreational	Recreational dominant	Balanced / Recreational	Recreational > Balanced
Health Impacts	Higher impacts than rural kids	Low impacts	High impacts across categories	Lower impacts than urban	Strong health issues (sleep, eye strain, anxiety)	Lower impacts than urban teens
Risk Profile	Moderate	Lower–moderate	High	Medium	Very high	Medium–high
Key Insight	High access + moderate use shapes early habits	Less access but similar hours	Device variety increases exposure	Same hours, fewer stress symptoms	Multi-device + late-night habits raise strain	Same hours without the heavy device stress